

SE3070 - Case Studies in Software Engineering

Assignment 02 - Proposed Improved Design

UC-04: Generate Post-Event Analysis Report

Improved version of the Group_026 design, Case Study 2:
Smart Disaster Early-Warning and Emergency Coordination System for Sri Lanka

Contents: 1. Use case scenario · 2. Use case diagram · 3. Class diagram · 4. Sequence diagram · 5. Wireframe changes · 6. Change justification

1. Corrected Use Case Scenario

Group_026 identified this use case in its use case diagram and summary but did not provide a scenario, sequence diagram, storyboard or wireframes. The design below is completed from their summary and diagram, and is consistent with the improved UC-01 and UC-03 designs, which supply its data.

Use Case ID	UC-04
Use Case Name	Generate Post-Event Analysis Report
Actors	Primary: DMC Officer. Secondary: Organisation (NGO / Donor) – receives shared reports.
Description	After a hazard event is closed, a DMC Officer generates a statistical report covering the alert timeline, citizens reached, shelter occupancy over time and resource distribution by district. Periods with missing data are flagged, and the report can be exported and shared with NGO or donor organisations.
Trigger	A hazard event has been closed, or an organisation requests an accountability report.
Preconditions	• The hazard event exists with status CLOSED. • Alert, notification, occupancy and distribution records exist for the event. • The DMC Officer is authenticated with reporting privileges.
Postconditions (success)	• A report is generated with all four sections and any data gaps flagged. • If exported, a ReportExport is stored. • If shared, a ReportShare is recorded with status SENT.
Postconditions (failure)	No file is produced and nothing is shared. Source data is never modified (the use case is read-only).

Main Flow

1. DMC Officer opens Reports → Post-Event Analysis.
2. System lists closed hazard events.
3. Officer selects an event; the system pre-fills its date range and affected districts.
4. Officer optionally narrows the date range or districts and selects Generate.
5. System validates the parameters (range within the event period, at least one district).
6. System compiles the alert timeline: each alert with time, hazard type, severity, scope, updates and all-clear.
7. System compiles citizens reached: unique citizens with at least one DELIVERED notification, and the delivery rate per channel.
8. System compiles shelter occupancy over time from occupancy records (daily peak per district).
9. System compiles resource distribution by district, supply type and owning organisation.
10. System checks each day of the range for missing records and marks gaps as “incomplete data” instead of omitting them.
11. System displays the report with summary figures and the four sections.
12. Officer exports the report as PDF or CSV.
13. System generates the file and confirms the export.
14. Officer shares the report with an organisation (selects the organisation and recipient email).
15. System sends the file, records the share and confirms it.
16. Use case ends.

Alternative Flows

A1 - Filter the report (step 11)

- A1.1 Officer filters by hazard type, district or organisation.
- A1.2 System recalculates and redisplay the report. Resume at step 12.

A2 - Export without sharing (step 14)

- A2.1 Officer downloads the file and ends the use case without sharing.

A3 - View only (step 12)

- A3.1 Officer closes the report without exporting. Nothing is stored except the generated report.

Exception Flows

E1 - Invalid parameters (step 5)

- E1.1 The date range is outside the event period, the start is after the end, or no district is selected.
- E1.2 System shows a validation error. Resume at step 4.

E2 - No data for the selection (steps 6-9)

- E2.1 No records exist for the selected range and districts.
- E2.2 System says no data is available. Resume at step 3.

E3 - Export failure (step 13)

- E3.1 The file cannot be generated.
- E3.2 System reports the failure; the report stays on screen and the officer can retry.

E4 - Sharing failure (step 15)

- E4.1 The file cannot be delivered.
- E4.2 System records the share as FAILED, keeps the exported file and lets the officer retry.

2. Corrected Use Case Diagram (UC-04 scope)

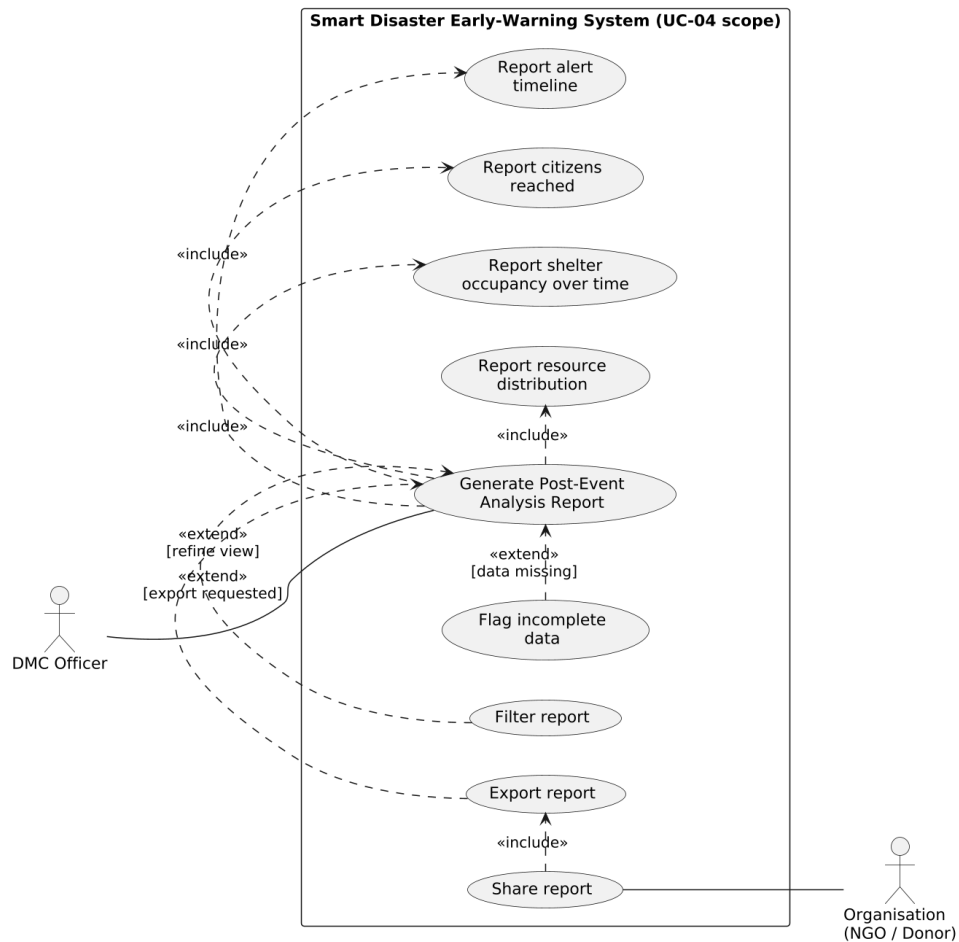


Figure 1: Improved use case diagram for Generate Post-Event Analysis Report

Changes from the original

- In the original diagram the District Officer was associated with this use case, but the summary says DMC officers and donor/NGO representatives. The primary actor is now DMC Officer.
- The Organisation actor was not connected to this use case, although donors receive reports. It is now associated with *Share report*.
- Kept the four report sections as «include»s, since every report contains them.
- Added «extend» *Flag incomplete data*, *Filter report* and *Export report*, and *Share report* which «include»s export. Export and sharing were missing from the original.

3. Corrected Class Diagram (reporting portion)

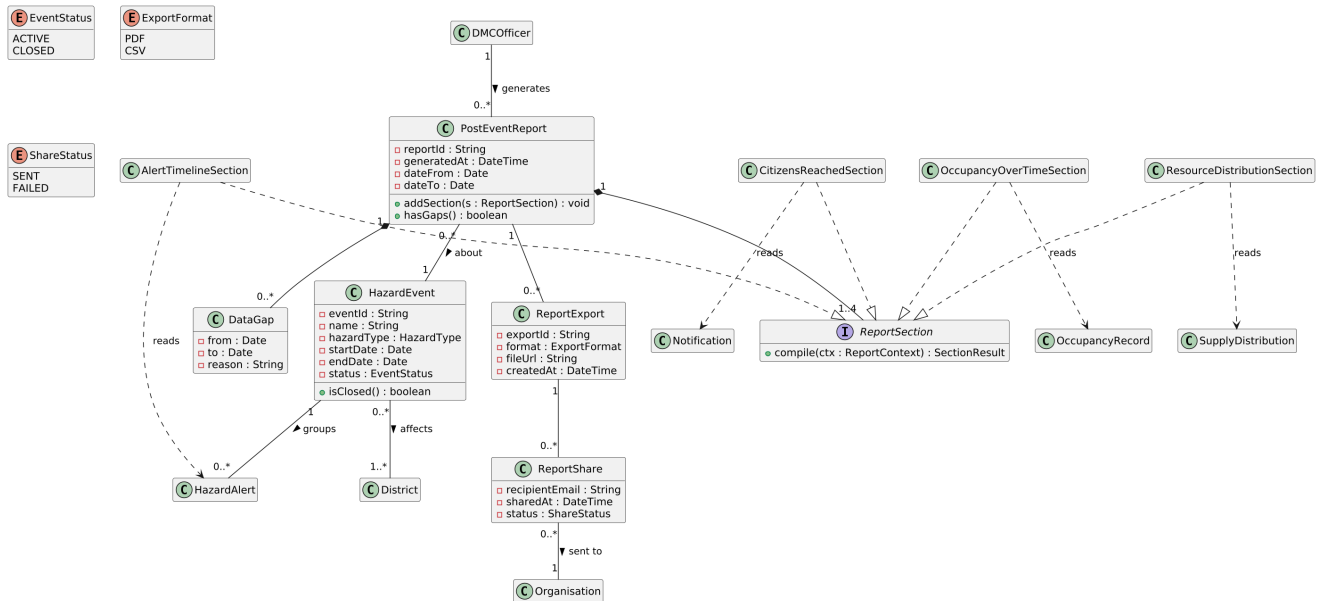


Figure 2: Improved class diagram for post-event reporting

Changes from the original

- Added HazardEvent to group alerts and affected districts. The original had no way to select “an event” to report on.
- Replaced the StatisticalReport class, which had four calculate methods, with a ReportSection interface and four implementations (Strategy pattern). Each section can be built and tested separately, and new sections can be added without changing existing code.
- Added DataGap, ReportExport and ReportShare for incomplete periods, files and sharing.
- Each section reads from the improved classes in UC-01 and UC-03: Notification (citizens reached), OccupancyRecord (occupancy over time) and SupplyDistribution (distribution). The original Shelter stored only current occupancy, which made “occupancy over time” impossible to calculate.

4. Sequence Diagram (new)

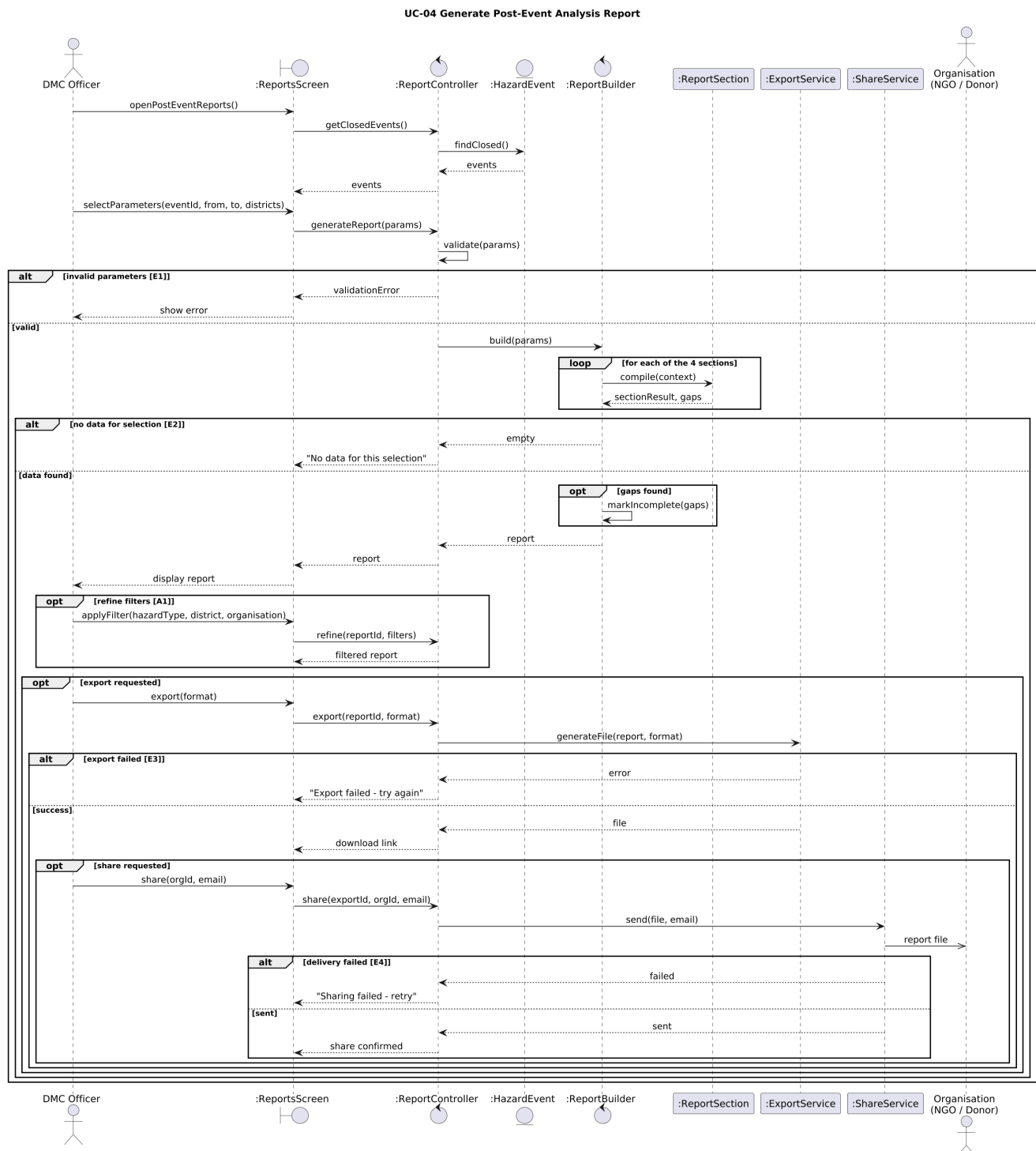


Figure 3: Sequence diagram – Generate Post-Event Analysis Report

Design notes

- The original design had no sequence diagram for this use case.
- ReportBuilder loops over the four ReportSection objects, matching the Strategy design in the class diagram.
- Gaps are marked, not dropped, so figures for incomplete periods are shown as partial.
- Covers filtering (A1), invalid parameters (E1), no data (E2), export failure (E3) and sharing failure (E4).

5. Wireframes (new)

5.1 Report parameters

DMC Command Console - Reports > Post-Event Analysis DMC Officer

Event

From To

Districts ☒ Colombo ☒ Gampaha ☒ Kalutara

Sections ☒ Alert timeline ☒ Citizens reached

☒ Shelter occupancy ☒ Resource distribution

Only closed events are listed. Dates must fall inside the event period.

Figure 4: Report parameter screen

- Uses the existing DMC Command Console layout; the original dashboard already had a “Reports” navigation item with no screen behind it.
- Only closed events are listed, and the date range is pre-filled from the event to prevent E1.

5.2 Report view and share dialog

Post-Event Report - Kelani basin floods - 3 districts

Filter:

(!) 14-15 Jun: incomplete data - figures partial, not omitted

Alerts issued	Citizens reached	Peak occupancy	Items distributed
14	128,400 (94%)	4,120	18,650

1. Alert timeline - markers by date and severity
2. Citizens reached - bars per alert, split by channel
3. Shelter occupancy over time - line per district, gap shaded
4. Resource distribution - grouped bars by district and organisation

Share report

Organisation

Recipient email

Format ☒ PDF ☐ CSV

Message

Figure 5: Post-event report view and share dialog

- Summary figures at the top, then the four sections, each with its chart type.
- A visible banner for incomplete data periods, so partial figures are never mistaken for complete ones.
- Filters for hazard type, district and organisation (A1), and Export and Share actions with a share dialog.

6. Change → Justification

Change	Critique it addresses
Full scenario, sequence diagram and wireframes	The original provided only a use case bubble and a one-line summary for this use case.
DMC Officer as primary actor; Organisation linked to sharing	The diagram linked District Officer, contradicting the summary, and did not connect donors to the report.
HazardEvent class	Reports are per event, but the original had no event concept to select or group alerts by.
ReportSection strategy	StatisticalReport held four unrelated calculations in one class (Single Responsibility and Open/Closed principles).
Uses OccupancyRecord and Notification from UC-03 / UC-01	The original data model could not produce occupancy over time or citizens reached.
Incomplete-data flagging	Offline periods are expected in disasters; silently omitting them would misrepresent the response.
Export and share with failure handling	The summary says reports go to donors, but no export or share function was designed.

Note for implementation and testing: each ReportSection can be unit-tested with small fixed data sets (for example, a citizen reached on two channels must be counted once), along with parameter validation, gap detection at range boundaries, and export and share failure paths.